Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (Benzene)
Basis set   : external-CI
Charge   : 0
Multiplicity : 1
Total e-   : 6
n-body order : 4

GROUND STATE ENERGIES

E0 (Ha) : nan
E1 (Ha) : nan
Gap (Ha) : nan
HF (Ha) : nan

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 20

C_LB : coherence witness bound
C_gcg: Conjugate Gradient(Alg. A)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 0
Frozen e-   : 0
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 0.777858
<D> / site : 0.129643

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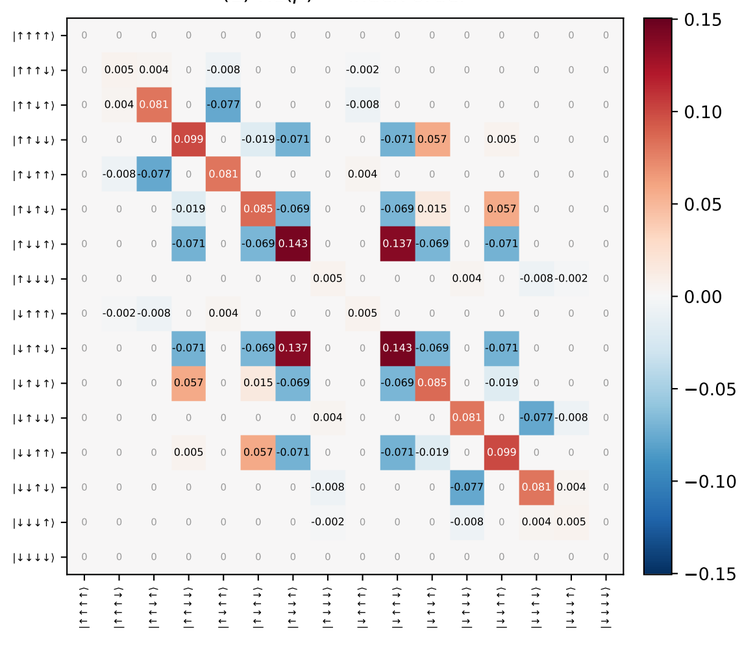
Body order : 4
Clusters   : 15

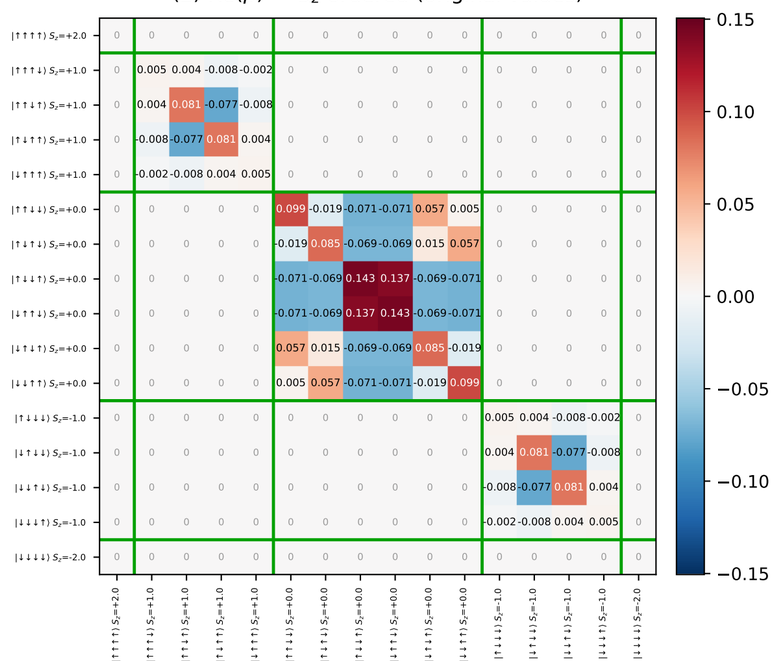
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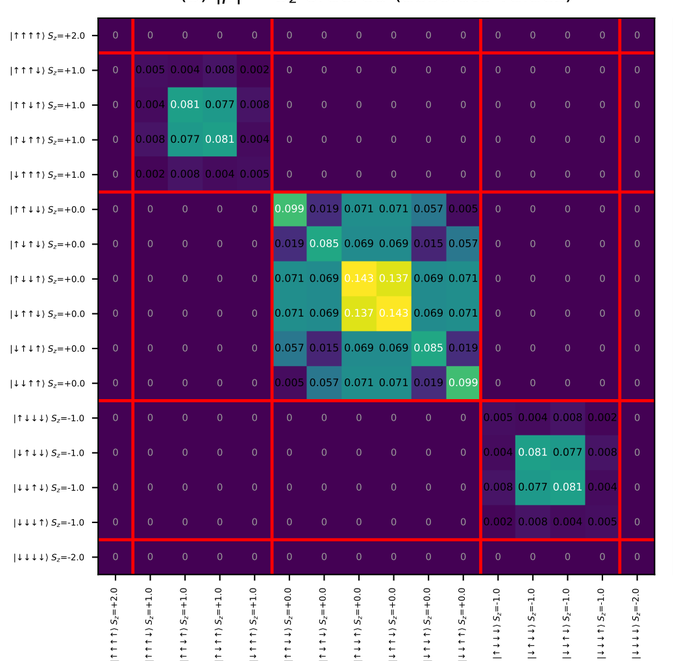
Summary — 4-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2,3	0.256423	1.614723	—	0.000000	0.135054	0.459631	MIXED
0,1,2,4	0.187748	1.920276	—	0.000000	0.016336	0.431185	STRONGLY MIXED
0,1,2,5	0.256423	1.614723	—	0.000000	0.130615	0.459631	MIXED
0,1,3,4	0.197011	1.840102	—	0.000000	0.316411	0.872831	STRONGLY MIXED
0,1,3,5	0.187748	1.920276	—	0.000000	0.029795	0.431185	STRONGLY MIXED
0,1,4,5	0.256423	1.614723	—	0.000000	0.133074	0.459631	MIXED
0,2,3,4	0.187748	1.920276	—	0.000000	0.024668	0.431185	STRONGLY MIXED
0,2,3,5	0.197011	1.840102	—	0.000000	0.316518	0.872831	STRONGLY MIXED
0,2,4,5	0.187748	1.920276	—	0.000000	0.023779	0.431185	STRONGLY MIXED
0,3,4,5	0.256423	1.614723	—	0.000000	0.129865	0.459631	MIXED
1,2,3,4	0.256423	1.614723	—	0.000000	0.134398	0.459631	MIXED
1,2,3,5	0.187748	1.920276	—	0.000000	0.024923	0.431185	STRONGLY MIXED
1,2,4,5	0.197011	1.840102	—	0.000000	0.316938	0.872831	STRONGLY MIXED
1,3,4,5	0.187748	1.920276	—	0.000000	0.027087	0.431185	STRONGLY MIXED
2,3,4,5	0.256423	1.614723	—	0.000000	0.130783	0.459631	MIXED

4-Body RDM (Sites 0,1,2,3) — same matrix in all three views



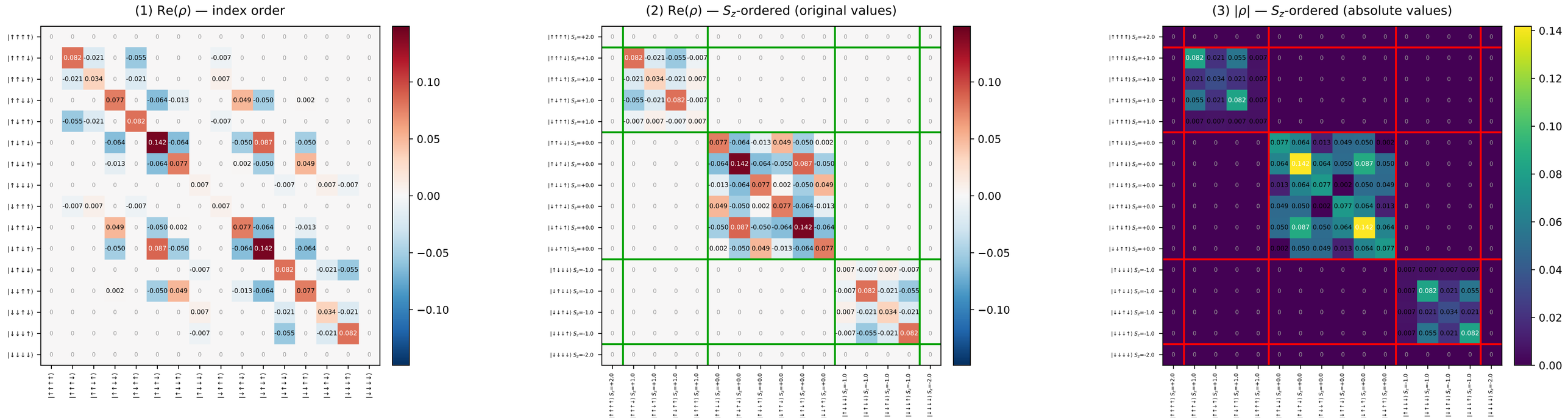




Trace=1.000000 Purity=0.256423 Entropy=1.614723 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.135054 ≤ C_{UB} = 0.459631 verdict: GME

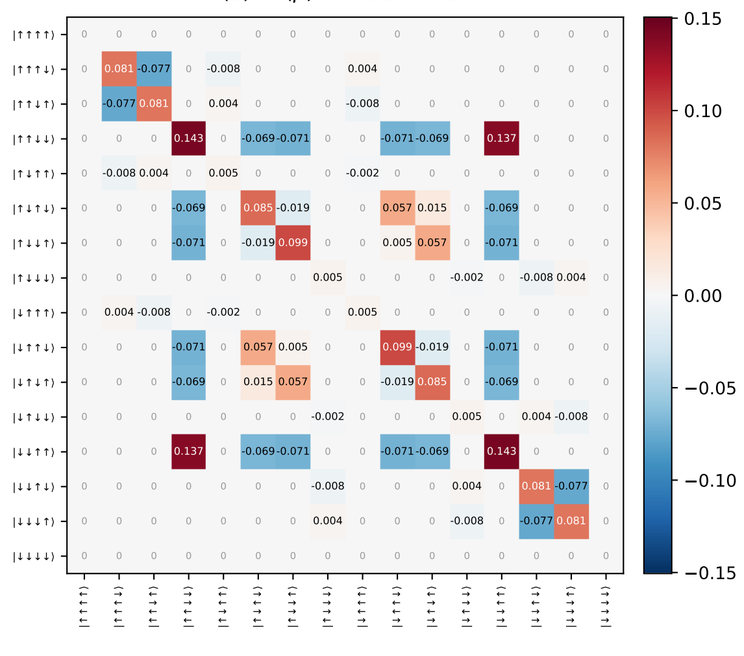
4-Body RDM (Sites 0,1,2,4) — same matrix in all three views

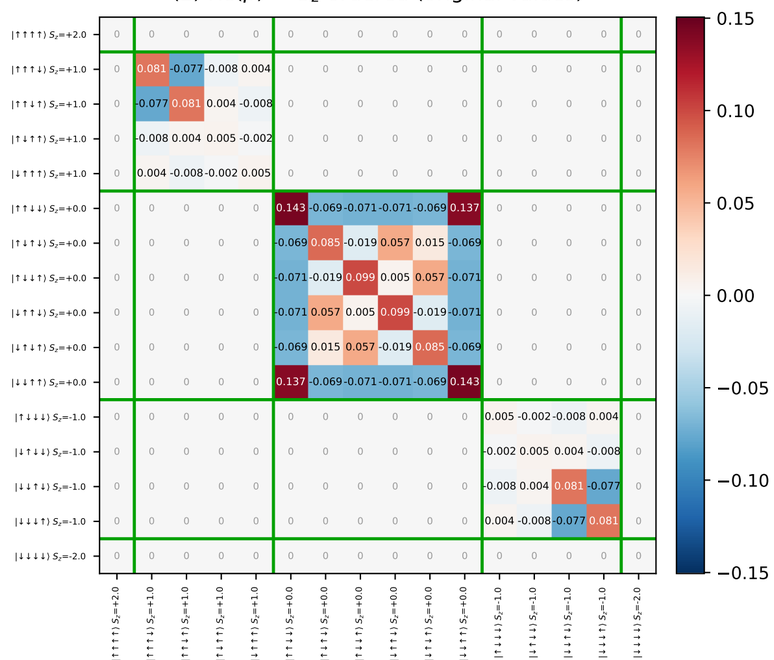


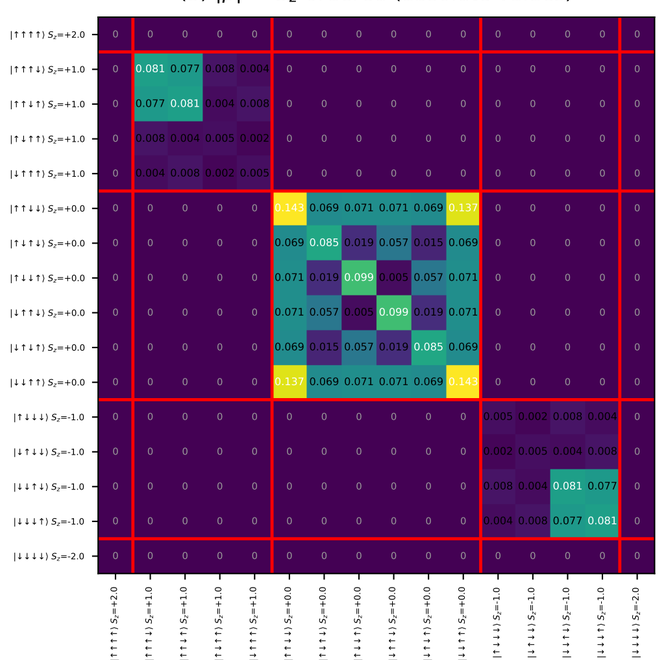
Trace=1.000000 Purity=0.187748 Entropy=1.920276 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.016336 ≤ C_{UB} = 0.431185 verdict: GME

4-Body RDM (Sites 0,1,2,5) — same matrix in all three views



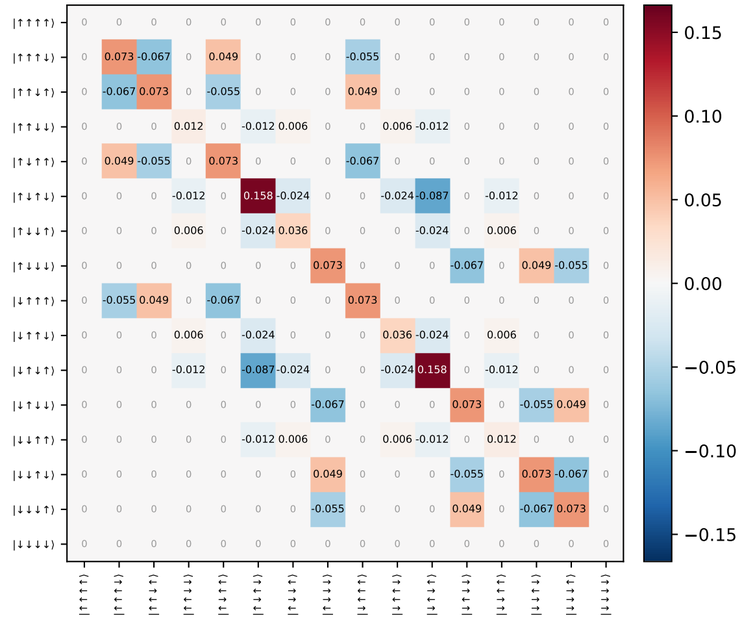


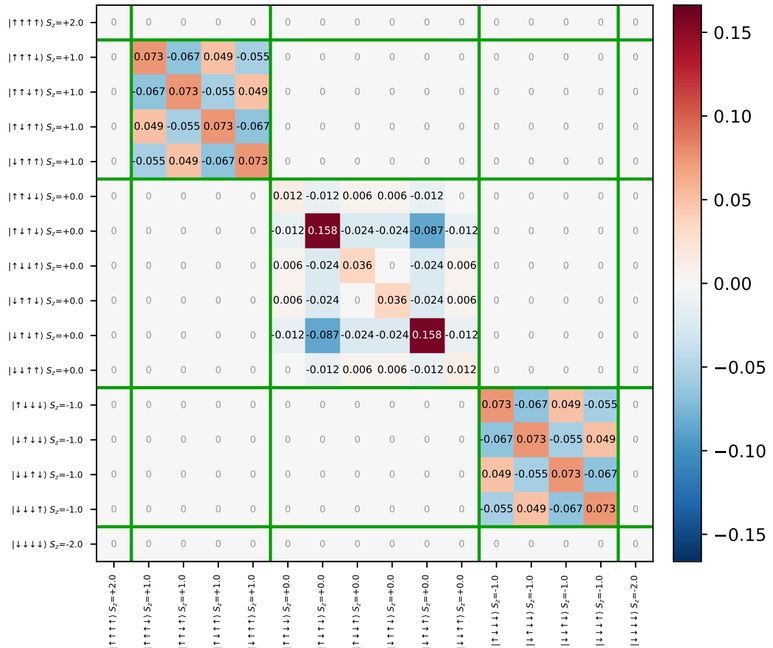


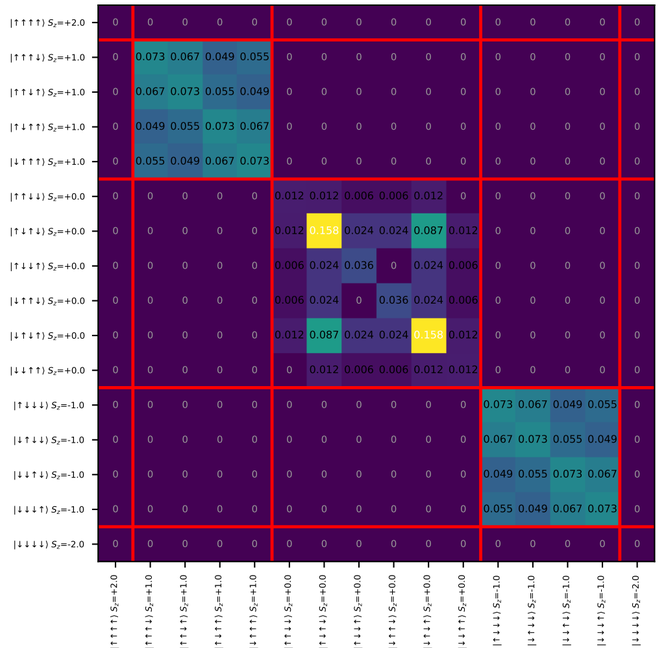
Trace=1.000000 Purity=0.256423 Entropy=1.614723 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.130615 ≤ C_{UB} = 0.459631 verdict: GME

4-Body RDM (Sites 0,1,3,4) — same matrix in all three views



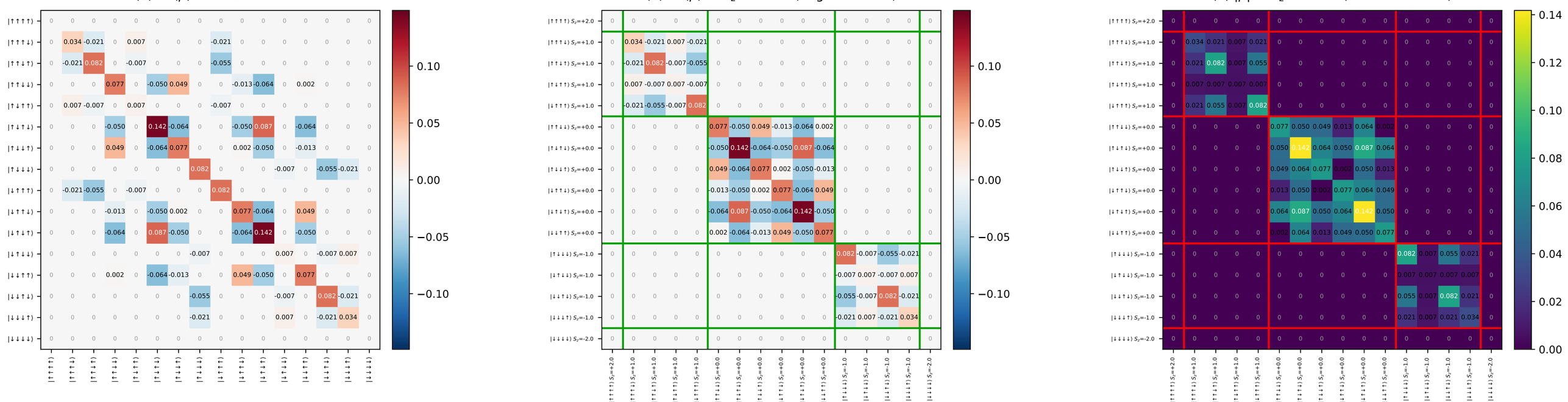




Trace=1.000000 Purity=0.197011 Entropy=1.840102 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.316411 ≤ C_{UB} = 0.872831 verdict: GME

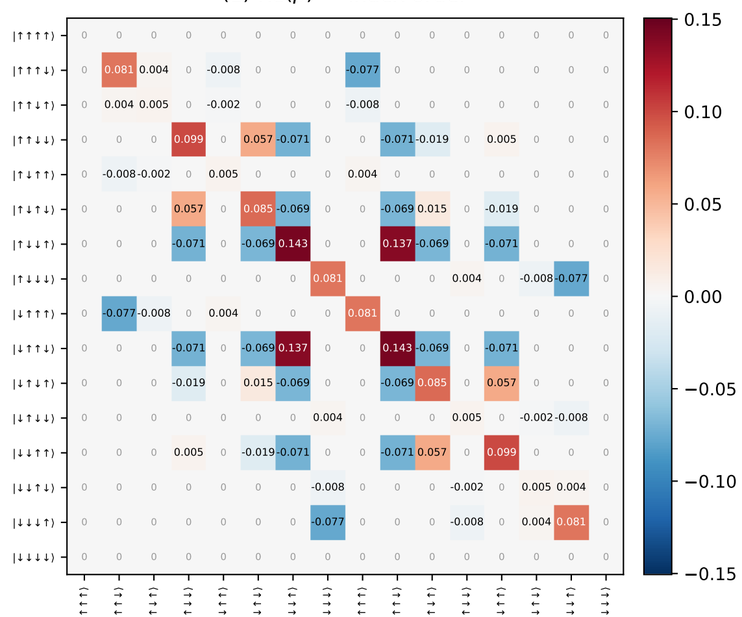
4-Body RDM (Sites 0,1,3,5) — same matrix in all three views

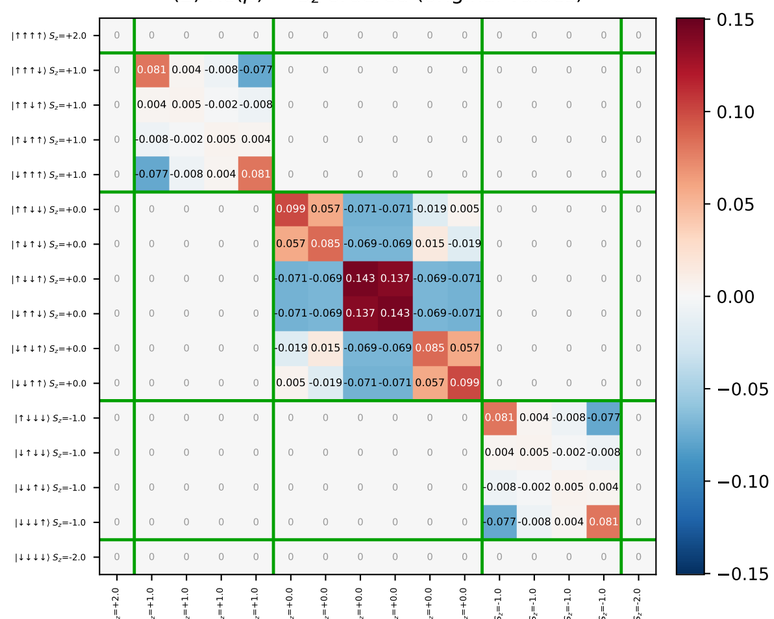


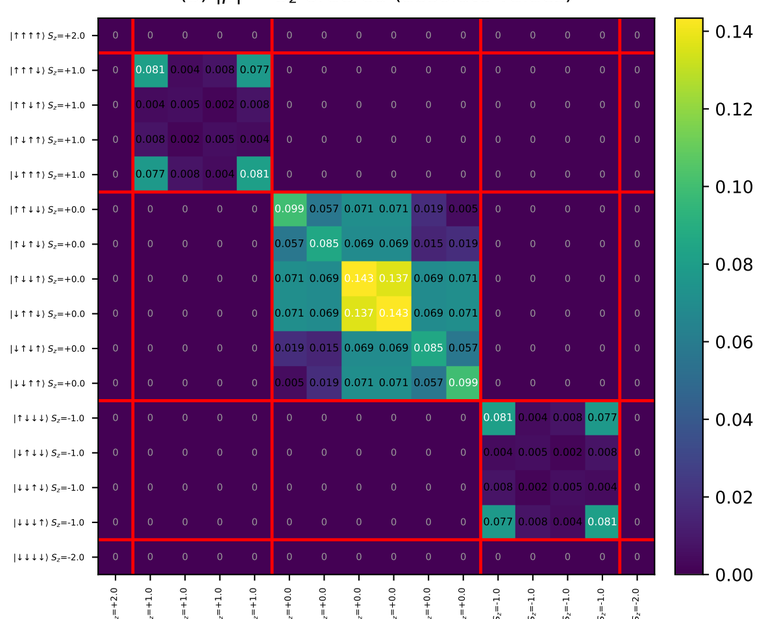
Trace=1.000000 Purity=0.187748 Entropy=1.920276 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.029795 ≤ C_{UB} = 0.431185 verdict: GME

4-Body RDM (Sites 0,1,4,5) — same matrix in all three views



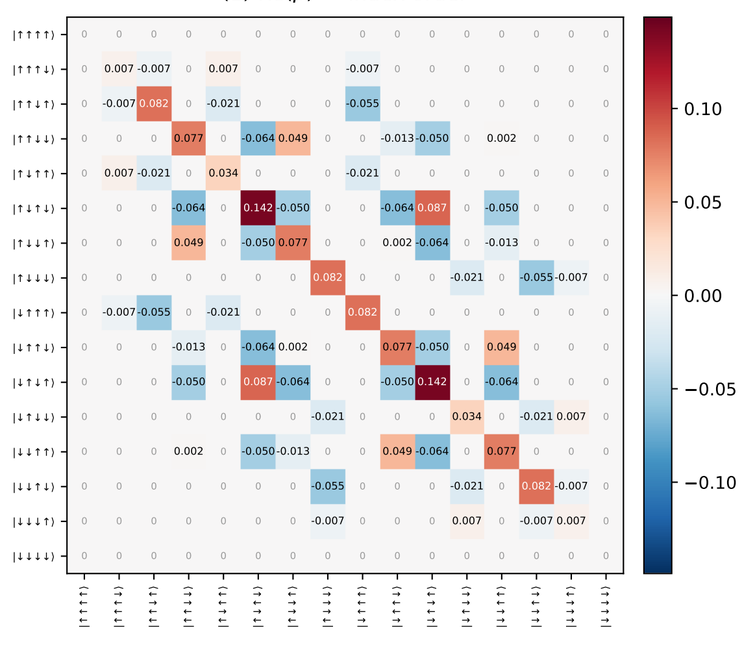


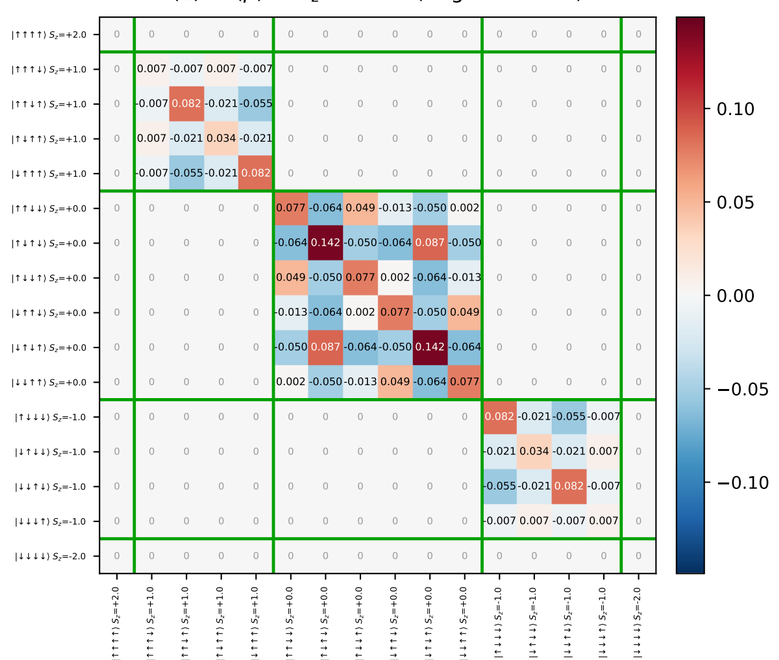


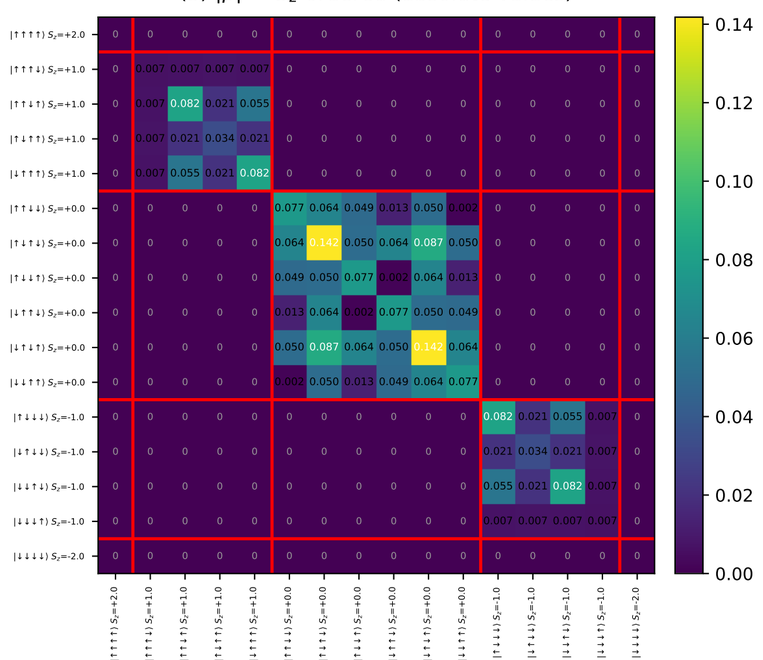
Trace=1.000000 Purity=0.256423 Entropy=1.614723 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.133074 ≤ C_{UB} = 0.459631 verdict: GME

4-Body RDM (Sites 0,2,3,4) — same matrix in all three views



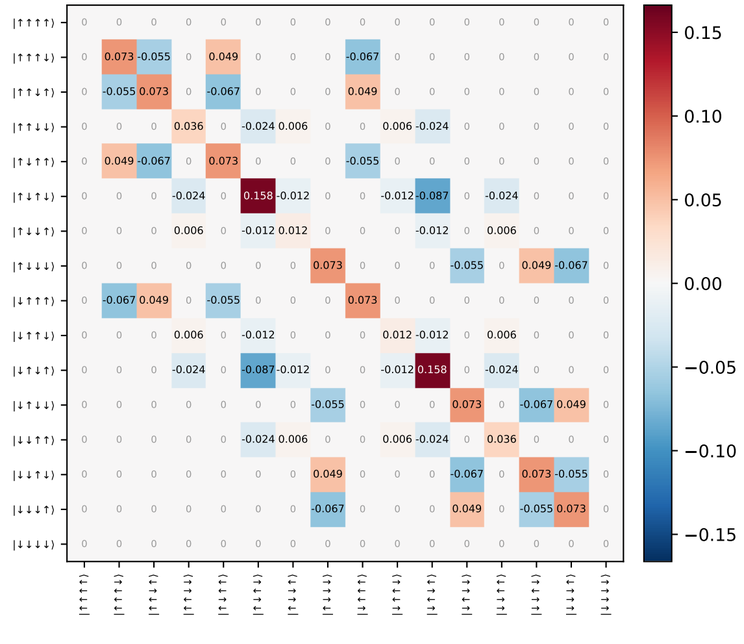


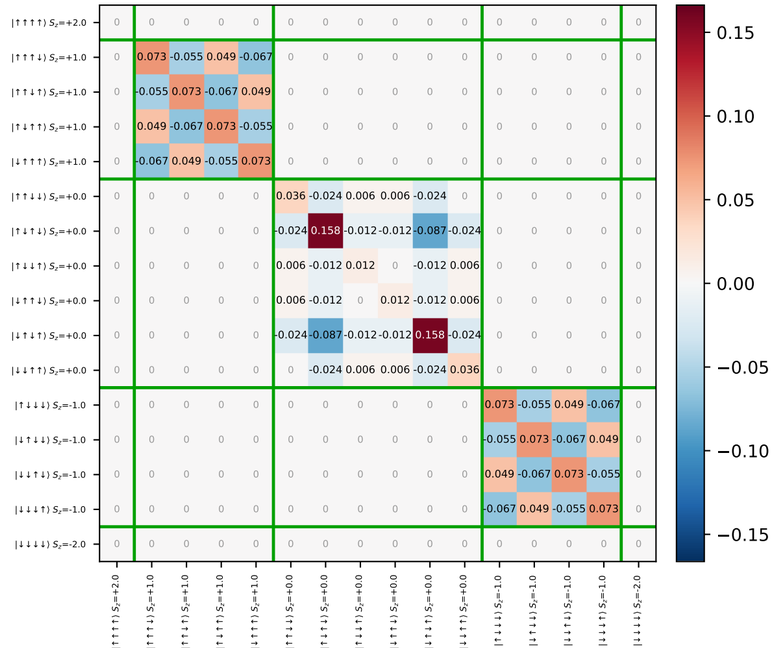


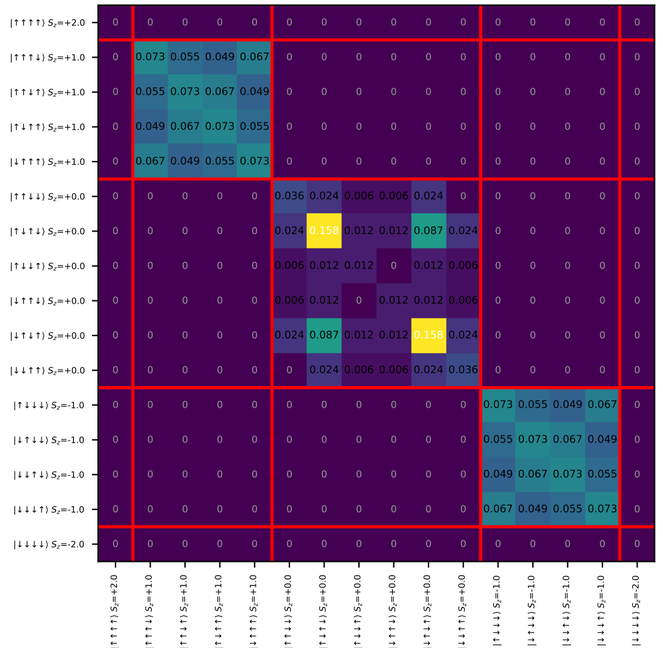
Trace=1.000000 Purity=0.187748 Entropy=1.920276 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.024668 ≤ C_{UB} = 0.431185 verdict: GME

4-Body RDM (Sites 0,2,3,5) — same matrix in all three views



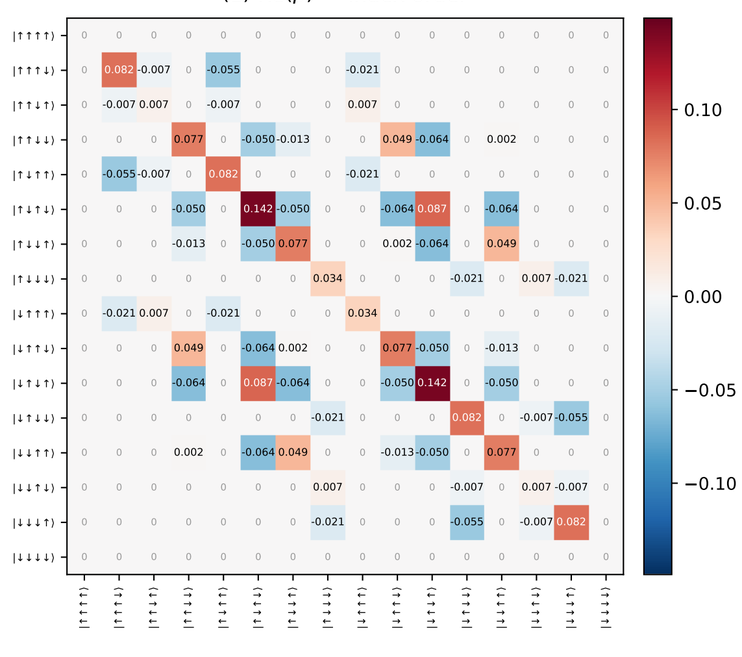


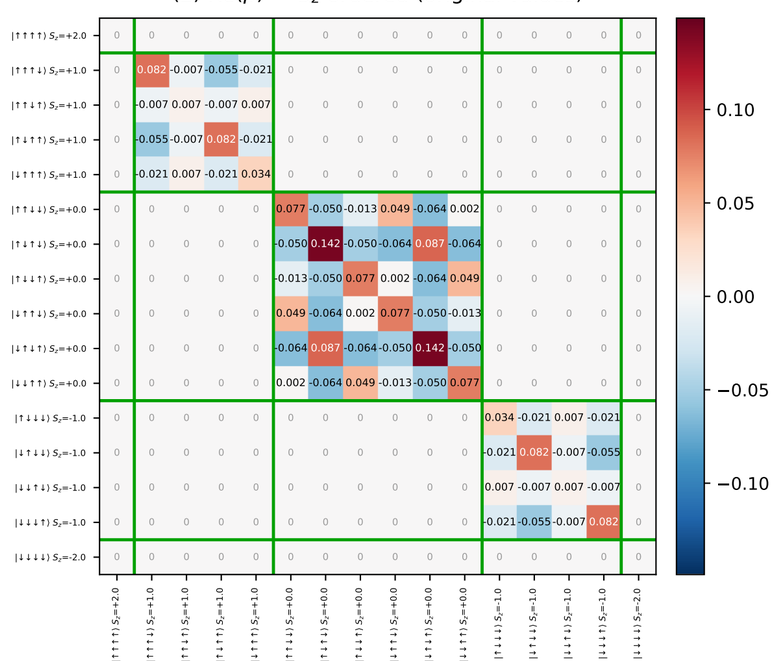


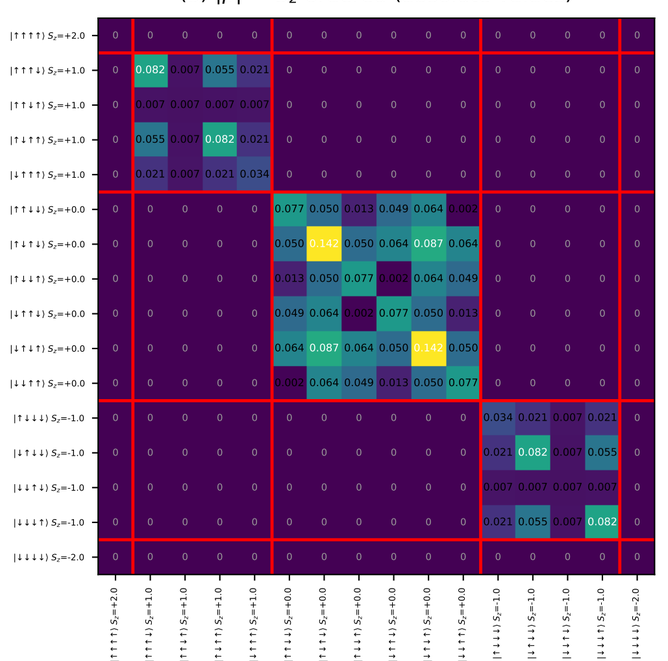
Trace=1.000000 Purity=0.197011 Entropy=1.840102 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.316518 ≤ C_{UB} = 0.872831 verdict: GME

4-Body RDM (Sites 0,2,4,5) — same matrix in all three views



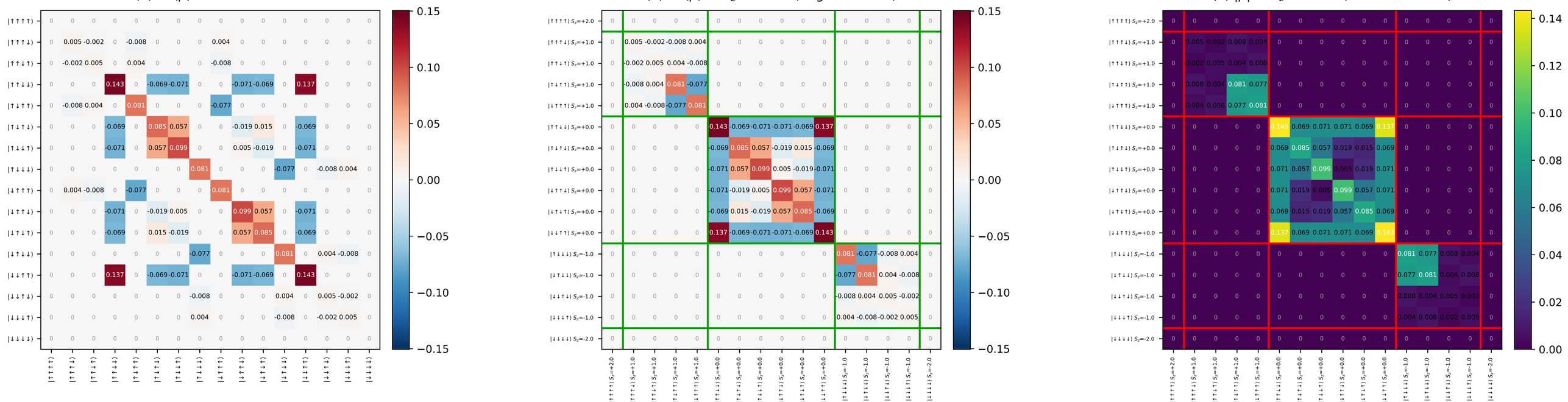




Trace=1.000000 Purity=0.187748 Entropy=1.920276 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.023779 ≤ C_{UB} = 0.431185 verdict: GME

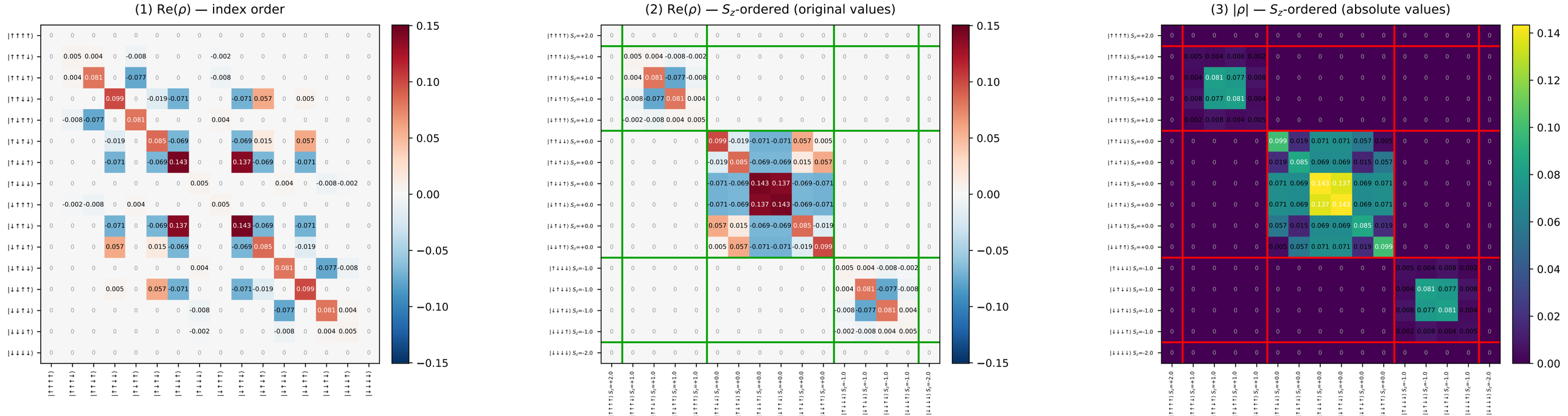
4-Body RDM (Sites 0,3,4,5) — same matrix in all three views



Trace=1.000000 Purity=0.256423 Entropy=1.614723 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

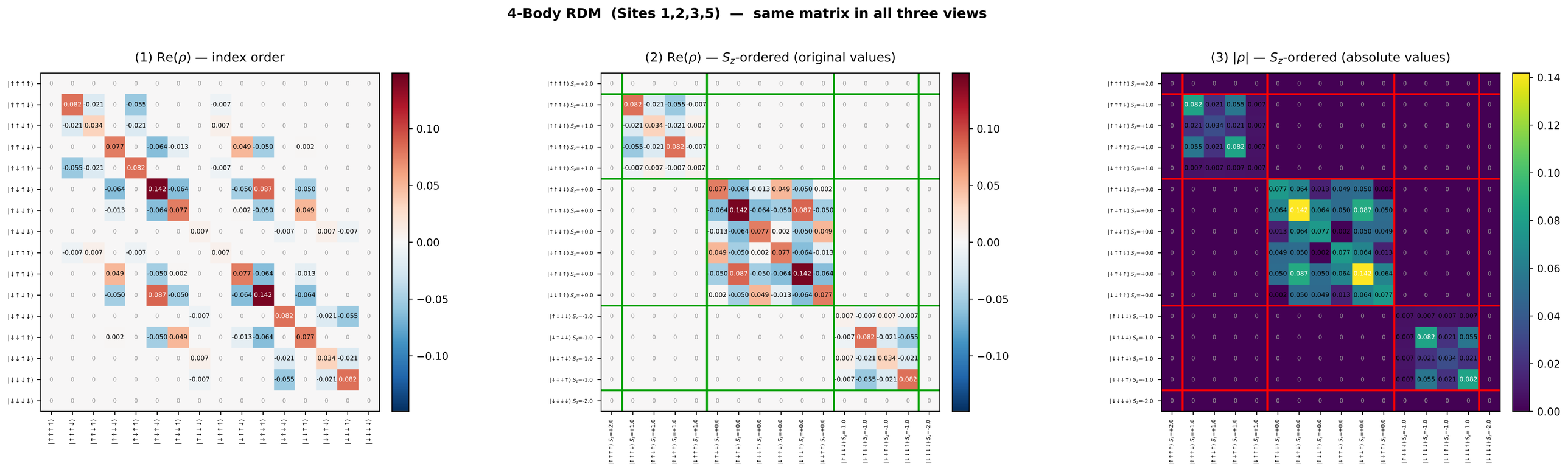
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.129865 ≤ C_{UB} = 0.459631 verdict: GME

4-Body RDM (Sites 1,2,3,4) — same matrix in all three views

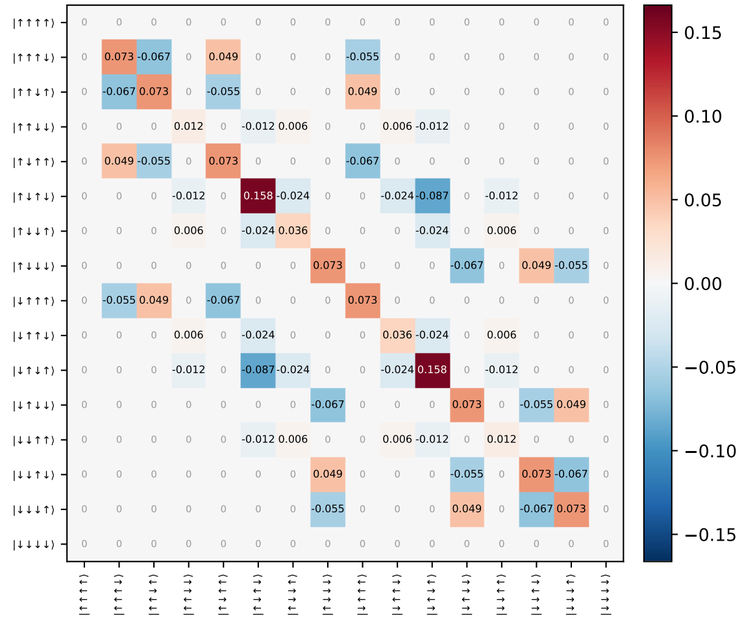


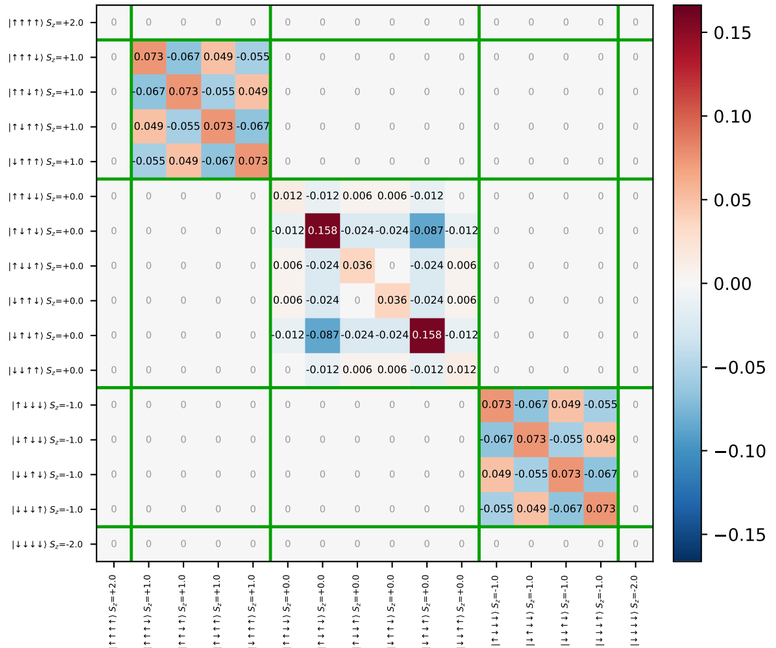
Trace=1.000000 Purity=0.256423 Entropy=1.614723 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

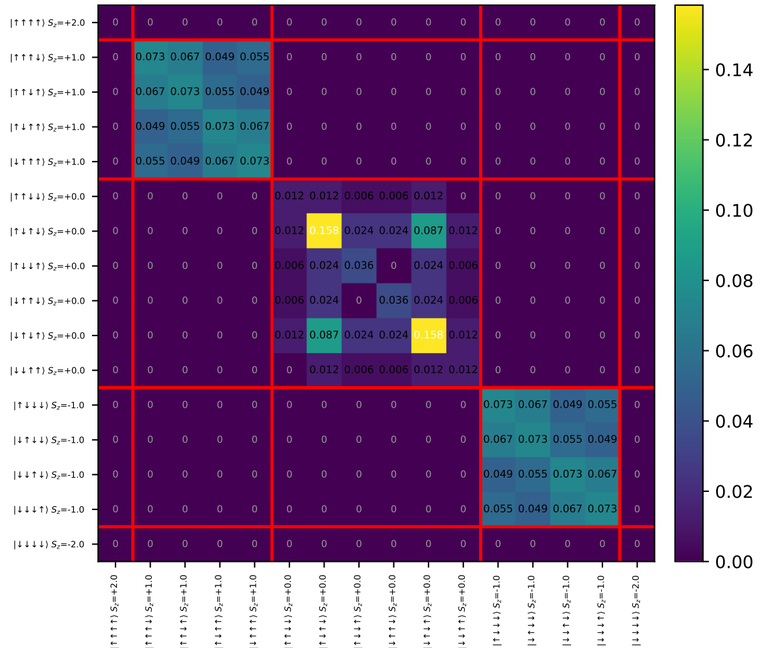
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.134398 ≤ C_{UB} = 0.459631 verdict: GME


4-Body RDM (Sites 1,2,3,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.187748 Entropy=1.920276 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.024923 ≤ C_{UB} = 0.431185 verdict: GME

4-Body RDM (Sites 1,2,4,5) — same matrix in all three views



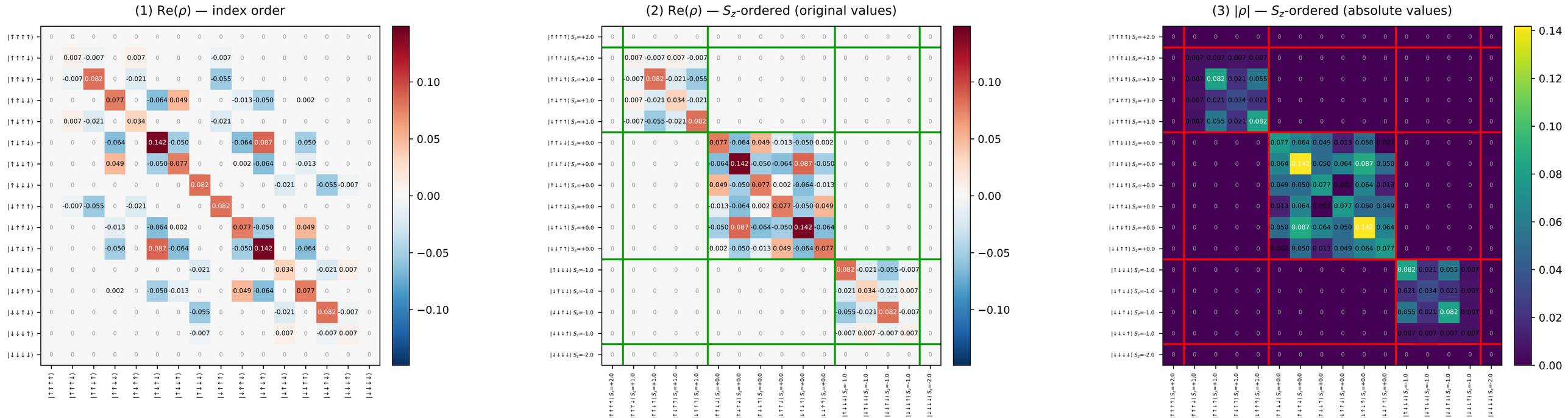




Trace=1.000000 Purity=0.197011 Entropy=1.840102 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.316938 ≤ C_{UB} = 0.872831 verdict: GME

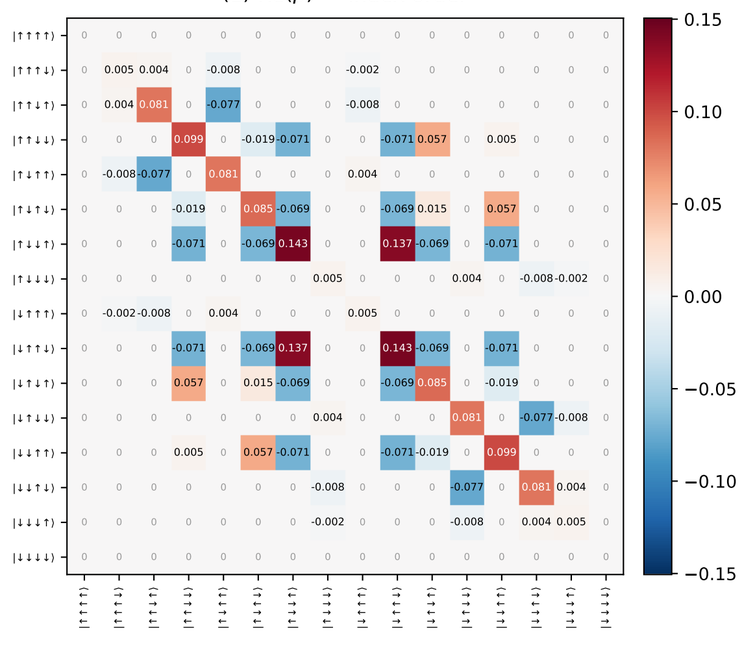
4-Body RDM (Sites 1,3,4,5) — same matrix in all three views

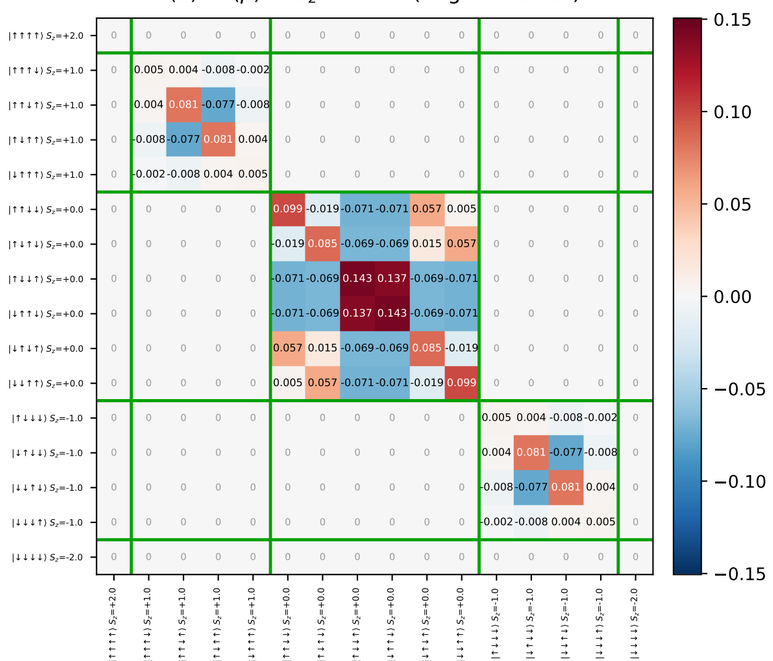


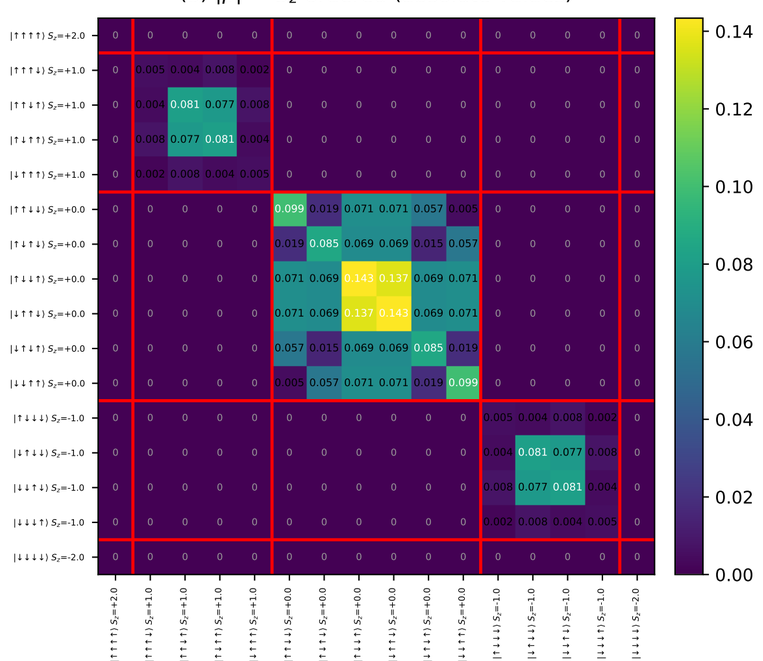
Trace=1.000000 Purity=0.187748 Entropy=1.920276 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.027087 ≤ C_{UB} = 0.431185 verdict: GME

4-Body RDM (Sites 2,3,4,5) — same matrix in all three views







Trace=1.000000 Purity=0.256423 Entropy=1.614723 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.130783 ≤ C_{UB} = 0.459631 verdict: GME